

Smart Home Dashboard - VS Code Setup Guide

Prerequisites

Before you start, make sure you have these installed on your computer:

1. **Node.js** (v18 or higher)
 - Download from: <https://nodejs.org/>
 - Verify installation: Open terminal/command prompt and run `node --version`
 2. **pnpm** (Package Manager)
 - Install globally: `npm install -g pnpm`
 - Verify installation: `pnpm --version`
 3. **VS Code** (Code Editor)
 - Download from: <https://code.visualstudio.com/>
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Step-by-Step Setup

Step 1: Extract the Project

1. Extract the `smart-home-dashboard.zip` file to a folder on your computer
2. You should see a folder named `smart-home-dashboard` with all the project files

Step 2: Open in VS Code

1. Open VS Code
2. Click **File** → **Open Folder**
3. Navigate to and select the `smart-home-dashboard` folder
4. Click **Open**

Step 3: Open Terminal in VS Code

1. In VS Code, press **Ctrl + `** (backtick) to open the integrated terminal
 - Or go to **View** → **Terminal**
2. You should see a terminal at the bottom of VS Code

Step 4: Install Dependencies

In the terminal, type the following command and press Enter:

```
Bash
```

```
pnpm install
```

This will download and install all required packages. This may take 2-5 minutes depending on your internet speed.

What to expect:

- You'll see many lines of output
- It should end with a message like "added X packages"
- A `node_modules` folder will be created

Step 5: Start the Development Server

Once installation is complete, type this command in the terminal:

```
Bash
```

```
pnpm dev
```

What to expect:

- You'll see output like:

```
Plain Text
```

```
→ Local:   http://localhost:3000/  
→ Network: http://169.254.0.21:3000/
```

- The app is now running locally

Step 6: Open in Browser

1. Open your web browser (Chrome, Edge, or Safari recommended for Web Speech API)
2. Go to: `http://localhost:3000`
3. You should see the Smart Home Dashboard with:
 - 5 device cards (Light, Fan, AC, TV, Main Door)
 - A voice command input bar at the bottom

- An activity log on the right side
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How to Use the Dashboard

Voice Commands

Click the **microphone icon** in the voice command bar to start listening. Try these commands:

- **"let there be light"** → Turns Light ON
- **"it is too bright"** → Turns Light OFF
- **"turn on the tv"** → Turns TV ON
- **"i am freezing"** → Turns AC OFF
- **"turn on fan"** → Turns Fan ON
- **"turn off fan"** → Turns Fan OFF
- **"lockdown"** → Locks the Main Door
- **"close door"** → Closes the Main Door

Manual Control

- Click on any device card to toggle it ON/OFF
- Click the toggle switch on each card for manual control
- Check the activity log on the right to see all actions

Mode Toggle

- If your browser doesn't support Web Speech API, you'll see "Simulation Mode"
 - Click the **Switch** button to toggle between Live Microphone and Simulation Mode
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Troubleshooting

Issue: "pnpm: command not found"

Solution: Install pnpm globally:

```
Bash
```

```
npm install -g pnpm
```

Issue: "Port 3000 is already in use"

Solution: Either:

1. Close the other application using port 3000
2. Or use a different port:

Bash

```
pnpm dev -- --port 3001
```

Issue: Microphone not working

Solution:

1. Make sure your browser is Chrome, Edge, or Safari (Firefox doesn't support Web Speech API)
2. Check that you've given the browser permission to access your microphone
3. Try the Simulation Mode by clicking the "Switch" button

Issue: Commands not being recognized

Solution:

1. Speak clearly and slowly
2. Make sure you're using the exact phrases listed above
3. Check the activity log to see what was transcribed
4. Try typing the command manually in the input field instead of using voice

Issue: App won't start after `pnpm dev`

Solution:

1. Stop the server: Press **Ctrl + C** in the terminal
 2. Clear cache: `pnpm store prune`
 3. Reinstall: `pnpm install`
 4. Start again: `pnpm dev`
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Building for Production

When you're ready to deploy:

Bash

```
pnpm build
```

This creates an optimized production build in the `dist/` folder.

To preview the production build locally:

Bash

```
pnpm preview
```

Project Structure

Plain Text

```
smart-home-dashboard/
├── client/                                # React frontend
│   ├── src/
│   │   ├── pages/
│   │   │   └── Home.tsx                # Main dashboard component
│   │   ├── hooks/
│   │   │   └── useSpeechRecognition.ts # Web Speech API hook
│   │   ├── components/                # UI components
│   │   ├── contexts/                  # React contexts
│   │   ├── index.css                  # Global styles & design tokens
│   │   └── main.tsx                   # React entry point
│   └── index.html                     # HTML template
├── server/                            # Express server
├── package.json                       # Project dependencies
├── vite.config.ts                     # Vite configuration
├── tsconfig.json                      # TypeScript configuration
└── README.md                          # Project documentation
```

Key Technologies Used

- **React 19** - UI framework
- **Tailwind CSS 4** - Styling

- **Framer Motion** - Animations
 - **Web Speech API** - Voice recognition
 - **TypeScript** - Type safety
 - **Vite** - Build tool
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Tips for Development

1. **Hot Reload:** Changes to your code automatically refresh the browser
 2. **Browser DevTools:** Press **F12** to open developer tools for debugging
 3. **Console Logs:** Check the browser console for any errors
 4. **Network Tab:** Use the Network tab to see API calls and performance
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Need Help?

If you encounter issues:

1. Check the browser console (F12) for error messages
 2. Check the terminal where you ran `pnpm dev` for server errors
 3. Make sure all prerequisites are installed correctly
 4. Try stopping the server and restarting with `pnpm dev`
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Next Steps

Once the app is running:

1. Test all voice commands
2. Try toggling devices manually
3. Check the activity log for all actions
4. Experiment with the simulation mode if Web Speech API isn't available
5. Customize the commands or add new devices as needed

Enjoy your Smart Home Dashboard! 🏠✨